

JavaScript vs TypeScript

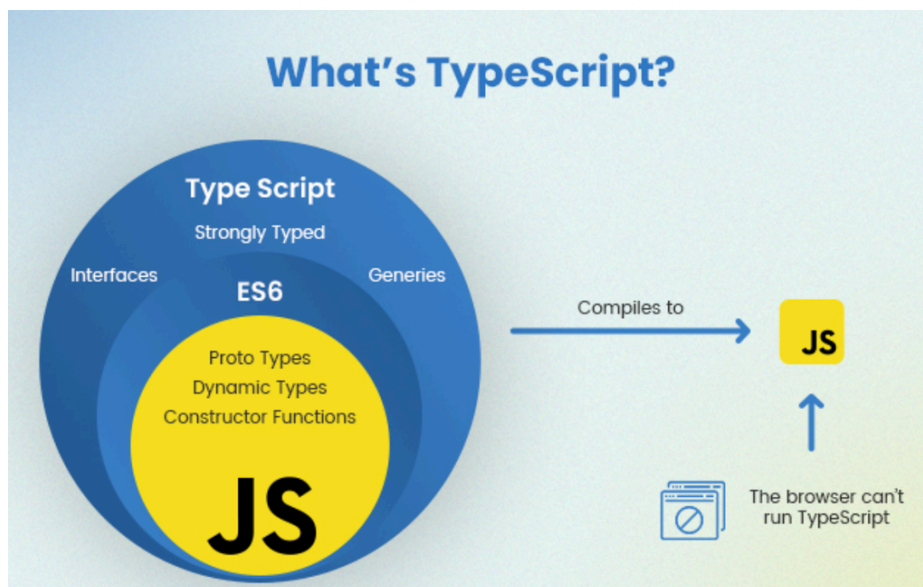
What is TypeScript?

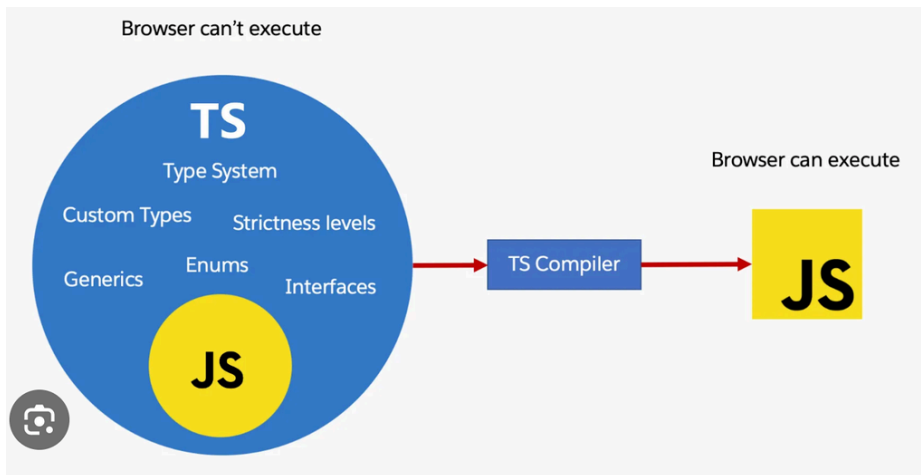
TypeScript (TS) is a **superset of JavaScript (JS)** developed by Microsoft.

That means:

- Every valid JavaScript program is also valid TypeScript.
- TypeScript adds **extra features** on top of JavaScript, the most important being **static typing**.
- It compiles (transpiles) down to plain JavaScript, which is what browsers actually run.

So, you write TypeScript → the TS compiler converts it into JS → browser/Node.js executes it.





JS	TS
JavaScript	TypeScript
Interpreted language	TS is a compiled language
Not trans-compiled-written in a single version	It is trans-compiled
Errors to get encountered during runtime	More errors during code writing
The learning curve is lower	The learning curve is higher
Less code to be written	Need to write more code
Can't predict how code will run	Codes are predictable
Remarkable Versatility	ECMA Script Features
Platform Independent	Use existing packages
Event-based Programming Features	Code editor support

JavaScript vs TypeScript

Feature	JavaScript	TypeScript
Type System	Dynamically typed[Runtime]	Statically typed with optional type annotations [Compile time]

Compilation	Interpreted directly by browsers or Node.js	Requires compilation/transpilation to JavaScript .ts-->tsc -->.js
Syntax	Standard ECMAScript syntax	Superset of JavaScript with additional syntax for types and interfaces
Error Checking	Errors found at runtime	Compile-time error checking
Tooling Support	Basic IDE support	Enhanced tooling with autocompletion, refactoring, and type checking
Interfaces & Types	No native support	Supports interfaces, enums, tuples, generics
Classes & OOP	ES6 classes supported	Supports ES6 classes plus advanced features like access modifiers (public/private/protected)
Modules	Supported via ES6 modules	Fully supports ES6 modules plus namespaces
Community & Ecosystem	Very large and mature	Growing rapidly; compatible with existing JS libraries but requires typings for full benefits

Learning Curve	Easier for beginners	Steeper due to static typing and additional concepts
Coding Speed	Faster prototyping due to dynamic typing. <code>.js</code> — <code>op</code>	Slower initially because of type annotations and compilation step <code>.ts --tsc --.js --run --op</code>
Prior Knowledge	Basic programming knowledge sufficient	Requires understanding of static typing concepts along with JavaScript fundamentals

Installation of typescript:

1. `npm install -g typescript`
2. `npm init -y`
3. `sudo npm install -D typescript`
4. `npx tsc -v`
5. `npx tsc --init`



package.json

```
{
  "name": "june2025typescriptsessions",
  "version": "1.0.0",
  "main": "index.js",
  "type": "module",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  }
}
```

```
  },  
  "keywords": [],  
  "author": "",  
  "license": "ISC",  
  "description": "",  
  "devDependencies": {  
    "typescript": "^5.9.2"  
  }  
}
```



tsconfig.json

```
{  
  "compilerOptions": {  
    "target": "es6",  
    "module": "nodenext",  
    "rootDir": "./src",  
    "outDir": "./dist",  
    "moduleResolution": "nodenext",  
    "esModuleInterop": true,  
    "skipLibCheck": true,  
    "strict": true,  
    "types": ["node"],  
    "noEmitOnError": true  
  },  
  "include": ["src/**/*.ts"],  
  "exclude": ["node_modules", "dist"]  
}
```

How to Install and Set Up TypeScript, Compile, and Run Your First .ts File

Follow these steps to install TypeScript, set it up in your project, compile a TypeScript file, and run it.

1. Install TypeScript Globally

```
1  npm install -g typescript
2
```

This installs the TypeScript compiler (tsc) globally on your system so you can use it anywhere.

2. Initialize a New Node.js Project

```
1  npm init -y
2
```

This creates a package.json file with default settings in your current directory.

3. Install TypeScript as a Development Dependency Locally

```
1  sudo npm install -D typescript
2
```

This installs TypeScript locally for your project (using sudo if needed for permissions).

4. Verify the Installed Version of TypeScript Compiler

```
1  npx tsc -v
2
```

This checks that the compiler is installed correctly by showing its version.

5. Create a tsconfig.json File to Configure the Project

```
1  npx tsc --init
```

```
2
```

This generates a `tsconfig.json` file where you can customize how TypeScript compiles your code.

Compile and Run Your First .ts File

1. **Create a new file named `hello.ts`**, for example:

```
console.log("Hello, TypeScript!");
```

2. **Compile the `.ts` file into JavaScript:**

```
npx tsc hello.ts
```

This will generate a `hello.js` JavaScript file in the same folder.

3. **Run the compiled JavaScript using Node.js:**

```
node hello.js
```

You should see this output:

```
1 Hello, TypeScript!  
2
```

Now you have successfully installed and set up TypeScript, compiled your first `.ts` file, and run it!